

Module 1

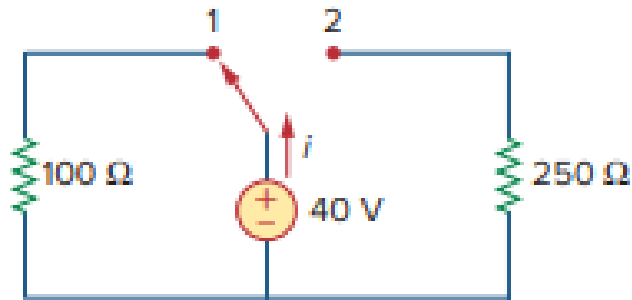
DC Circuit Analysis and Theorems

Dr. Elizabeth Rufus

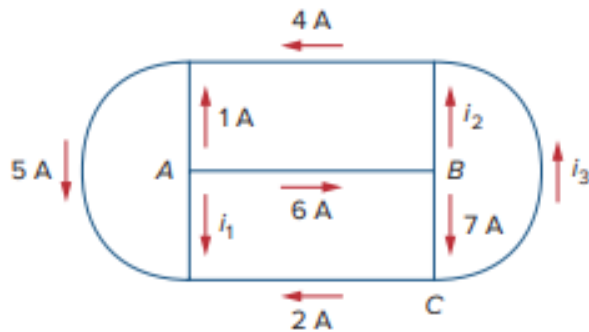
Professor, School of Electronics Engineering

Tutorial

1. Find the current I when the switch is in position 1 and then when in position 2.

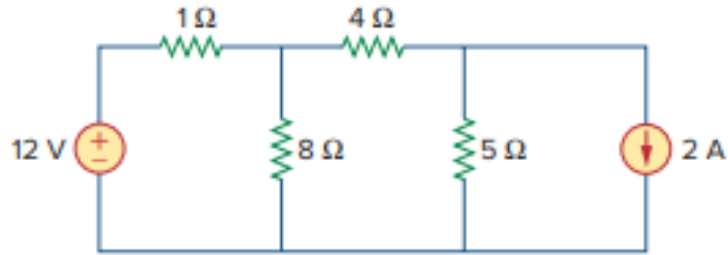


2. Find the currents i_1 , i_2 , i_3 .

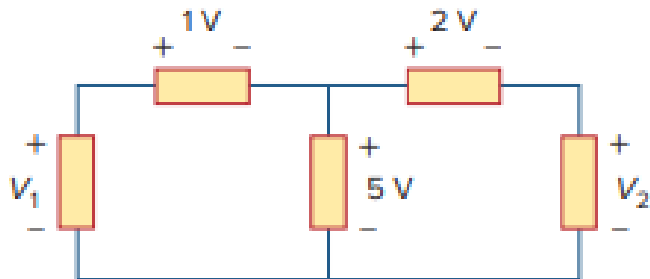


Tutorial

3. Determine the number of branches and nodes in the circuit given below.

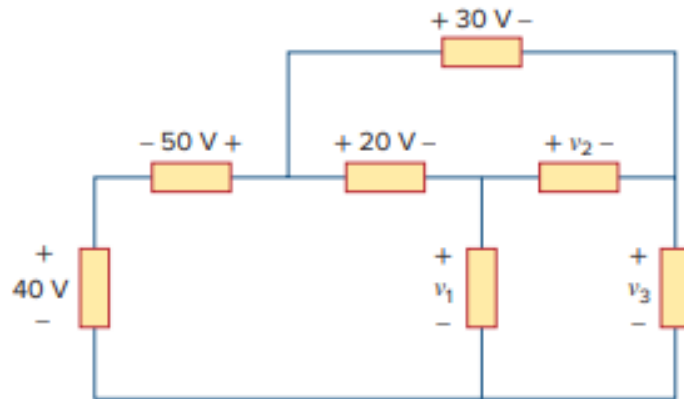


4. Calculate V_1 and V_2 .

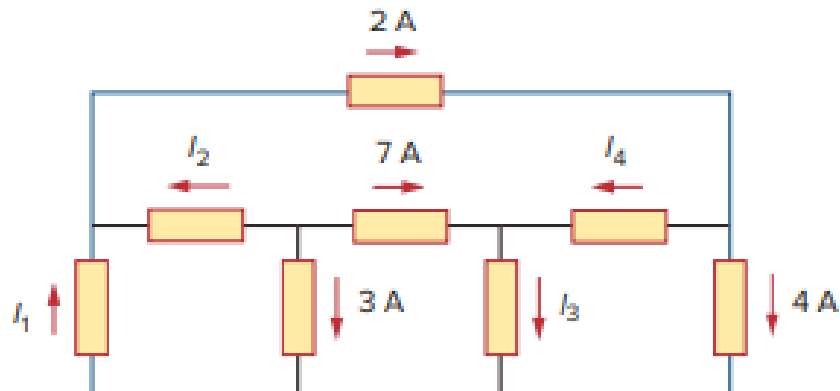


Tutorial

5. Calculate V_1 , V_2 and V_3

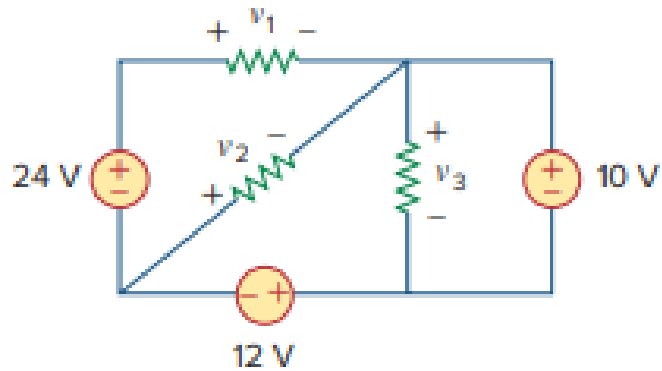


6. Find currents i_1 to i_4 .

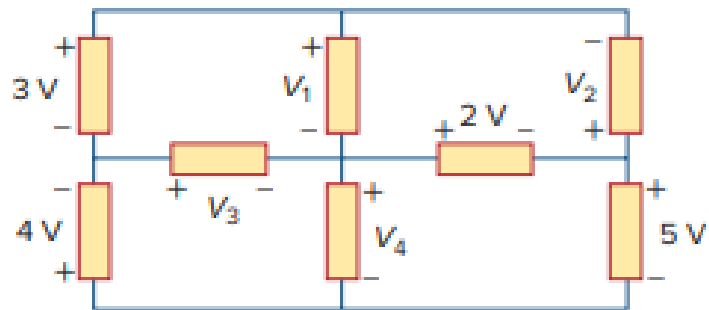


Tutorial

7. Determine V_1 through V_3 in the circuit.

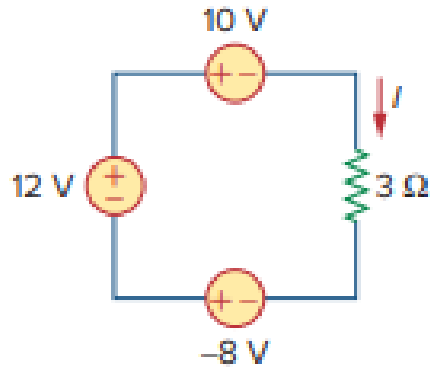


8. Calculate V_1 to V_4 using KVL.

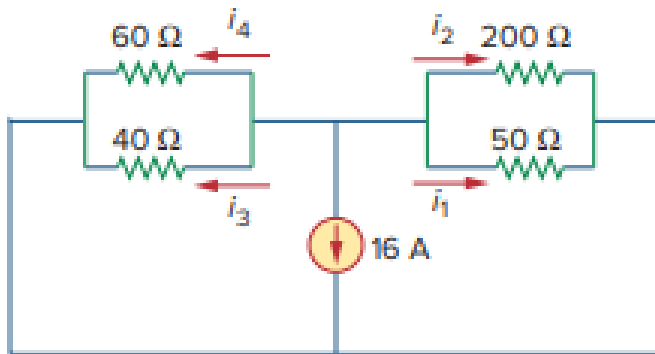


Tutorial

9. Find I , the power dissipated by the resistor, and the power supplied by each source.

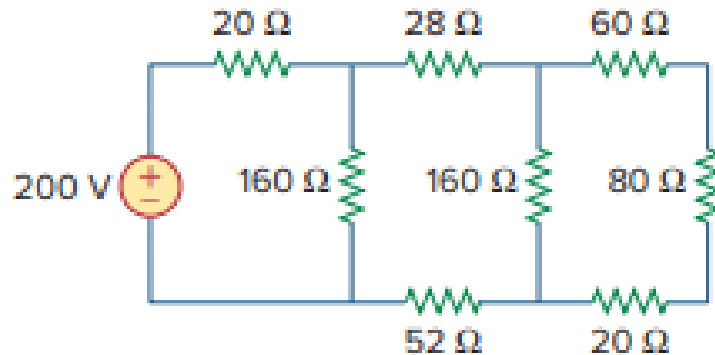


10. Calculate i_1 to i_4 in the circuit.

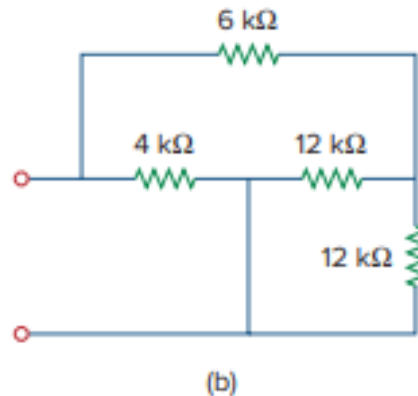
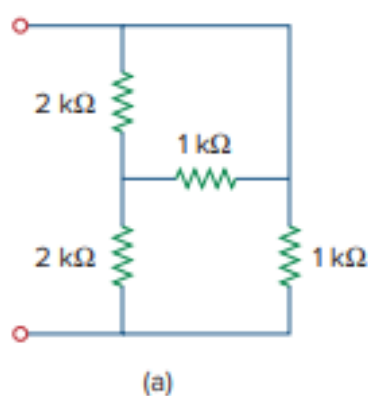


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11. Using series/parallel resistance combination, find the equivalent resistance seen by the source in the circuit. Find the overall absorbed power by the resistor network.

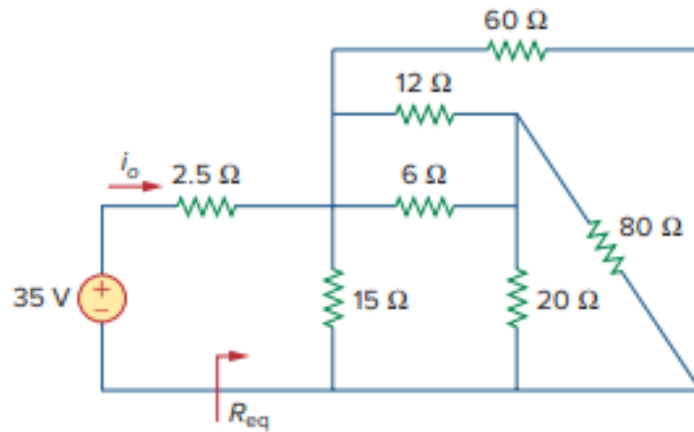


12. Evaluate R_{eq} looking into each set of terminals for each of the circuits.

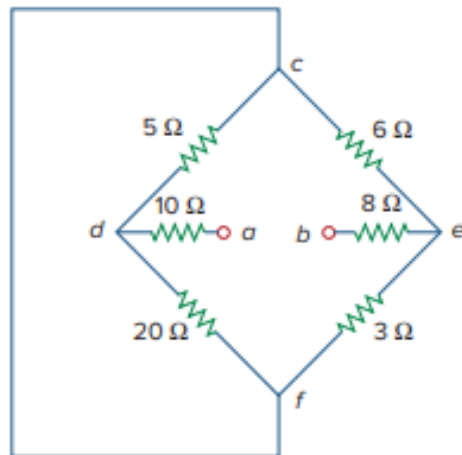


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13. Find R_{eq} and i_o in the circuit.

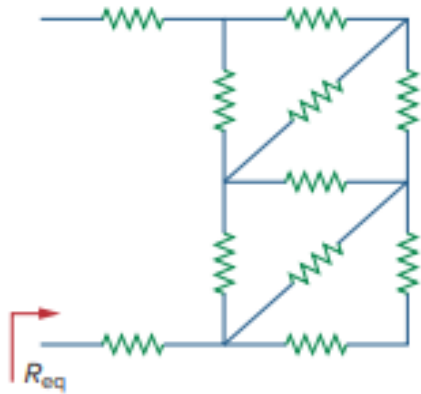


14. Find the equivalent resistance R_{ab} in the circuit.

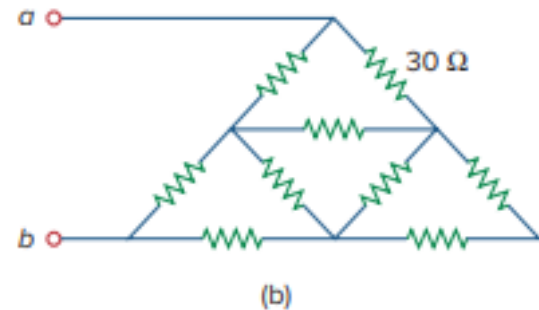
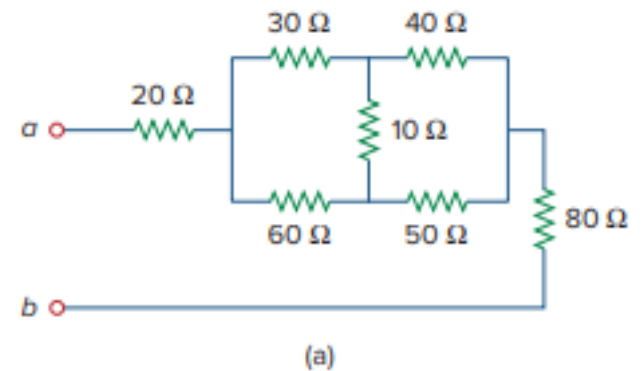


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15. For the circuit shown below, find the equivalent resistance. All resistors are 3Ω .

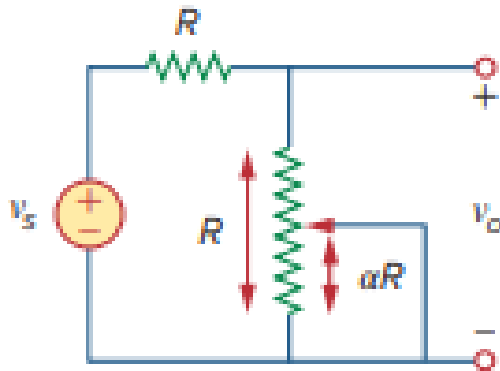


16. Obtain the equivalent resistance R_{ab} in each of the circuits (a) and (b). In (b), all resistors have a value of 30Ω .



Tutorial

17. In the circuit, the wiper divides the potentiometer resistance between αR and $(1 - \alpha)R$, $0 \leq \alpha \leq 1$. Find v_o / v_s .



18. Two delicate devices are rated as shown. Find the values of the resistors R_1 and R_2 needed to power the devices using a 24-V battery.

